



Technical Reproducibility Validation Report

STRaitRazor

Version v3.0

STR / SNP panel

Verification date	2026-07-06
Repository commit	b618e9345ab40f348b504083ae8de2b39abb60fa
Attestation level	Runs + Output Structure Validation
Permalink	https://strhub.app/verified/straitrazor-v3-0
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution — confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

- Source available
- Installation successful
- End-to-end execution
- Output generated

NOT VERIFIED

- Genotype accuracy
- Concordance
- Forensic validity
- Regulatory compliance



1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	5/5 gates passed — Runs + Output Structure Validation
ATTESTATION LEVEL	Runs + Output Structure Validation
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Reproducibility Metadata

Tool	STRaitRazor
Version	v3.0
Repository	https://github.com/Ahhgust/STRaitRazor
Commit (pinned)	b618e9345ab40f348b504083ae8de2b39abb60fa
Container OS	ubuntu-22.04
Marker panel	STR / SNP panel
Verified on	2026-07-06 20:11:43 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/28820084645

3. Exact Run Command

The following command was executed verbatim in the CI environment. Replacing the input paths under /data/in with your own inputs reproduces the verified run.

```
# Environment: ubuntu-22.04 · STRaitRazor v3.0 · commit b618e93
$ /opt/tool/str8rzt \
-c \
/opt/tool/ForenSeqv1.27.config \
/data/in/sample.fastq \
> \
/data/out/sample.allsequences.txt
```

4. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS

Output Structure Validation	Output structure matches expected genotype-bearing format	PASS
Execution log	—	

5. Out of Scope

- This report does not evaluate any of the following:
- Genotype correctness or accuracy
 - Concordance against known truth sets
 - Sensitivity, specificity, or stutter performance
 - Allele calling accuracy or forensic casework suitability
 - Regulatory compliance or ISO accreditation
 - Multi-laboratory or multi-dataset reproducibility

6. Output Content Evidence

Output file	sample.allsequences.txt
Format	TSV
Sequence records	816
Distinct STR loci	63
Total reads	3,967
Max depth (single locus)	132

7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below.

Dataset	NIST mds2-2157 — Illumina STR (ForenSeq slice, donor NTD01)
Source	https://data.nist.gov/od/id/mds2-2157
License	research / training / education only (per NIST); not for donor identification or database searching
Ref. genome	—
Loci tested	Amelogenin · CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D17S1301 · D18S51 · D19S433 · D1S1656 · D20S482 · D21S11 · D22S1045 · D2S1338 · D2S441 · D3S1358 · D4S2408 · D5S818 · D6S1043 · D7S820 · D8S1179 · D9S1122 · DXS10074 · DXS10103 · DXS10135 · DXS7132 · DXS7423 · DXS8378 · DYF387S1 · DYS19 · DYS385 · DYS389I · DYS389II · DYS390 · DYS391 · DYS392 · DYS437 · DYS438 · DYS439 · DYS448 · DYS460 · DYS461 · DYS481 · DYS505 · DYS522 · DYS533 · DYS549 · DYS570 · DYS576 · DYS612 · DYS635 · DYS643 · FGA · HPRTB · N29insA · PentaD · PentaE · TH01 · TPOX · Y-GATA-H4 · mh16-MC1RB · mh16-MC1RC · vWA

8. Verification Matrix

PASS	Reference dataset	NIST mds2-2157 — Illumina STR (ForenSeq slice, donor NTD01)
	README check (5/5)	Advisory — all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

9. Limitations

- Single reference dataset per input type
- Single containerized environment (Docker / ubuntu-22.04)
- No truth-set comparison or ground-truth genotypes
- No accuracy or concordance assessment
- No forensic validation of results
- Short-read limitations apply (very long STR alleles may not span reads)

10. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

11. Conclusion

◆ Runs end-to-end

STRaitRazor v3.0 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 5 verification gates were passed.

◆ Produces valid output

The tool generated a structurally valid TSV output file with genotype calls across 63 target forensic STR loci, with a maximum read depth of 132 at DYS392.

◆ No bundled demo data

STRaitRazor does not include its own test or demo dataset. STRhub Verified supplied NIST mds2-2157 — Illumina STR (ForenSeq slice, donor NTD01) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ Reproducibility statement

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.



Appendix — Detected Markers with Read Depth

STRaitRazor v3.0 · verified 2026-07-06 · showing the 60 deepest of 220 detected markers by read depth

